Continuity in Type Theory

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A short introduction to Martin-Löf type theory

The Curry-Howard correspondence

First order logic		Dependent type theory	
$\bot$	$\cong$	$\mathbb{0}$	
$\top$	$\cong$	$\mathbb{1}$	
$P \wedge Q$	$\cong$	$P \times Q$	
$P \vee Q$	$\cong$	$P + Q$	
$P \Rightarrow Q$	$\cong$	$P \to Q$	
$\forall(x:A).P(x)$	$\cong$	$\Pi(x:A).P(x)$	(dependent function)
$\exists(x:A).P(x)$	$\cong$	$\Sigma(x:A).P(x)$	(dependent pair)

Axiom of choice?

$$\Pi(x:X).\Sigma(y:Y).A(x,y) \to \Sigma(f:X \to Y).\Pi(x:X).A(x,fx)$$

Brouwer's continuity principle

The value of $f(\alpha)$ of a function $f : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}$ depends only on a finite prefix of the sequence $\alpha \in \mathbb{N}^{\mathbb{N}}$.

In *e.g.* higher-type Heyting arithmetic (HA^{ω}),

$$\forall(f : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}).\ \forall(\alpha : \mathbb{N}^{\mathbb{N}}).\ \exists(n : \mathbb{N}).\ \forall(\beta : \mathbb{N}^{\mathbb{N}}).\ \alpha =_n \beta \Rightarrow f(\alpha) = f(\beta)$$

is not provable (or disprovable).

But it's validated by *e.g.* Johnstone's topological topos, among other well-known models.

Continuity in type theory

How should it be formulated in intuitionistic type theory?

- We of course don't expect it to be provable.
- But much less we expect it to be disprovable.

Its Curry-Howard interpretation (CH - Cont)

$$\Pi(f : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}).\ \Pi(\alpha : \mathbb{N}^{\mathbb{N}}).\ \Sigma(n : \mathbb{N}).\ \Pi(\beta : \mathbb{N}^{\mathbb{N}}).\ \alpha =_n \beta \to f(\alpha) = f(\beta)$$

is provably false in intensional Martin-Löf type theory.

What does it mean?

What is the the correct formulation of the continuity principle in type theory?

What about uniform continuity of functions $\mathbf{2}^{\mathbb{N}} \to \mathbb{N}$?

Failure of the CH interpretation of the continuity principle

A theorem in intensional Martin-Löf type theory (with $\mathbb{N}, \Pi, \Sigma, \mathrm{Id}$):

$$\Big(\Pi(f\!:\!\mathbb{N}^{\mathbb{N}}\!\to\!\mathbb{N})(\alpha\!:\!\mathbb{N}^{\mathbb{N}}).\ \Sigma(n\!:\!\mathbb{N}).\ \Pi(\beta\!:\!\mathbb{N}^{\mathbb{N}}).\ \alpha =_n \beta \to f(\alpha) = f(\beta)\Big) \to 0 = 1$$

by adaptation of an old argument due to Kreisel, originally relying on choice and extensionality.

By projection, $\mathrm{CH}\text{-}\mathrm{Cont}$ gives a modulus-of-continuity functional

$$M : (\mathbb{N}^{\mathbb{N}} \to \mathbb{N}) \times \mathbb{N}^{\mathbb{N}} \to \mathbb{N}$$

assigning a modulus $n = M(f, \alpha)$ to f at the point α.

Trouble: While all functions $\mathbb{N}^{\mathbb{N}} \to \mathbb{N}$ may be continuous, there can't be any continuous modulus-of-continuity functional.

Proof of $\mathrm{CH}\text{-}\mathrm{Cont} \to 0 = 1$

- Assuming $\mathrm{CH}\text{-}\mathrm{Cont}$, we get M and write $M(f) = M(f, 0^\omega)$, where
  - 0^ω is the infinite sequence of zeros, and
  - $0^n k^\omega$ consists of n zeros followed by infinitely many k's.
  - Two facts: $0^\omega =_n 0^n k^\omega$ and $(0^n k^\omega)(n) = k$ for any n, k.
- Let $m = M(\lambda\alpha.0)$.

  Define $f : \mathbb{N}^{\mathbb{N}} \to \mathbb{N}$ to be $f(\beta) = M(\lambda\alpha.\beta(\alpha m))$.
- By expanding the definitions (which involves the ξ-rule), we get
$$f(0^\omega) = M(\lambda\alpha.0^\omega(\alpha m)) = M(\lambda\alpha.0) = m.$$
- By the definition of M, we have
$$\Pi(\beta : \mathbb{N}^{\mathbb{N}}).\ 0^\omega =_{M(f)} \beta \to m = f\beta.$$
- Choosing $\beta = 0^{M(f)+1}1^\omega$, we have $0^\omega =_{M(f)} \beta$ and hence $f(\beta) = m$.
- By the continuity of $\lambda\alpha.\beta(\alpha m)$, we get
$$\Pi(\alpha : \mathbb{N}^{\mathbb{N}}).\ 0^\omega =_m \alpha \to \beta 0 = \beta(\alpha m).$$
- Choosing $\alpha = 0^m(M(f)+1)^\omega$, we have $0^\omega =_m \alpha$ and hence
$$0 = \beta 0 = \beta(\alpha m) = \beta(M(f)+1) = 1.$$

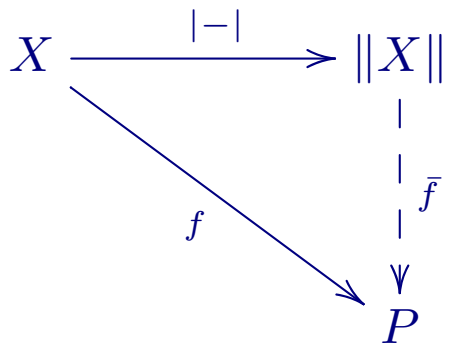
Discussion

1. No continuous modulus-of-continuity functional M:
   We used our hypothetical M to define a non-continuous function f and hence prove M wrong.
2. And this is exactly what is happening in the topological topos:
   - All functions $\mathbb{N}^{\mathbb{N}} \to \mathbb{N}$ are continuous.
   - But there is no continuous way of finding moduli of continuity.
3. The conversion
   $$f(0^\omega) = M(\lambda\alpha.0^\omega(\alpha m)) = M(\lambda\alpha.0) = m$$
   in the proof relies on the ξ-rule (reduction under λ).
4. In HA^ω, the ξ-rule, the axiom of choice, and the continuity of all functions $\mathbb{N}^{\mathbb{N}} \to \mathbb{N}$ are together impossible.
5. Since ξ-rule holds in categories, any locally cartesian closed category with a natural numbers object (*e.g.* any topos) disproves $\mathrm{CH}\text{-}\mathrm{Cont}$.

Propositional truncation

A type is called a proposition if it has at most one element.

A propositional truncation of a type X, if it exists, is a proposition $\|X\|$ together with a map $|-| : X \to \|X\|$ such that for any proposition P and $f : X \to P$ we can find $\bar{f} : \|X\| \to P$.



Intuitively, $\|X\|$ is

- the truth value of the inhabitedness of X;
- the quotient of the type X by the chaotic equivalence relation.

The logic of propositions in HoTT and toposes

In HoTT, $\|X\|$ is defined as a higher inductive type.

In a topos, $\|X\|$ is the image of the unique map $X \to 1$.

The logic of propositions

$$\begin{aligned}
\bot &:\equiv \mathbb{0} \\
\top &:\equiv \mathbb{1} \\
P \wedge Q &:\equiv P \times Q \\
P \vee Q &:\equiv \|P + Q\| \\
P \Rightarrow Q &:\equiv P \to Q \\
\forall(x:A).P(x) &:\equiv \Pi(x:A).P(x) \\
\exists(x:A).P(x) &:\equiv \|\Sigma(x:A).P(x)\|
\end{aligned}$$

The correct type-theoretic formulation of continuity

$$\Pi(f:\mathbb{N}^{\mathbb{N}}\to\mathbb{N})(\alpha:\mathbb{N}^{\mathbb{N}}).\ \|\ \Sigma(n:\mathbb{N}).\ \Pi(\beta:\mathbb{N}^{\mathbb{N}}).\ \alpha =_n \beta \to f(\alpha) = f(\beta)\ \|$$

- It's validated in *e.g.* the topological topos.
- The continuous dependency of n on inputs f and α is now broken.
- Because the axiom of choice

  $$\Pi(x:X).\|\Sigma(y:Y).A(x,y)\| \to \|\Sigma(f:X\to Y).\Pi(x:X).A(x,y)\|$$

  is not provable.

Uniform continuity of functions $\mathbf{2}^{\mathbb{N}} \to \mathbb{N}$

$$\forall(f : \mathbf{2}^{\mathbb{N}} \to \mathbb{N}).\ \exists(n : \mathbb{N}).\ \forall(\alpha, \beta : \mathbf{2}^{\mathbb{N}}).\ \alpha =_n \beta \Rightarrow f(\alpha) = f(\beta)$$

- Not provable but consistent in HA^{ω},
  validated by the topological topos (using classical logic or other axioms).
- Its Curry–Howard interpretation
  $$\Pi(f : \mathbf{2}^{\mathbb{N}} \to \mathbb{N}).\ \Sigma(n : \mathbb{N}).\ \Pi(\alpha, \beta : \mathbf{2}^{\mathbb{N}}).\ \alpha =_n \beta \to f(\alpha) = f(\beta)$$
  is also consistent in MLTT.
- Moreover, it's logically equivalent to
  $$\Pi(f : \mathbf{2}^{\mathbb{N}} \to \mathbb{N}).\ \| \ \Sigma(n : \mathbb{N}).\ \Pi(\alpha, \beta : \mathbf{2}^{\mathbb{N}}).\ \alpha =_n \beta \to f(\alpha) = f(\beta)\ \|$$
  assuming function extensionality.
- We have a constructive variation of the topological topos.

Disclosing secrets from truncations

In general we don't have $\|X\| \to X$ for arbitrary X, because it gives a constructive taboo and also contradicts univalence.

However, for some types X, we can disclose a secret $\|X\|$ to X.

Lemma. For any family A of types indexed by natural numbers such that

1. $A(n)$ is a proposition for every $n : \mathbb{N}$, and
2. $A(n)$ implies that $A(m)$ is decidable for every $m < n$

we have

$$\Sigma(n{:}\mathbb{N}).A(n) \;\leftrightarrow\; \|\Sigma(n{:}\mathbb{N}).A(n)\|.$$

Proof sketch of ($\leftarrow$). Given n with $A(n)$, we can find the minimal k with $A(k)$, using the decidability of $A(m)$ for $m < n$. Since "having a minimal k with $A(k)$" is a proposition (proved using function extensionality), the elimination rule of $\| - \|$ gives the desired result.

Equivalence of the two formulations of uniform continuity

Theorem. The proposition

$$\Pi(f : \mathbf{2}^{\mathbb{N}} \to \mathbb{N}).\ \| \ \Sigma(n : \mathbb{N}).\ \Pi(\alpha, \beta : \mathbf{2}^{\mathbb{N}}).\ \alpha =_n \beta \to f(\alpha) = f(\beta) \ \|$$

is logically equivalent to the type

$$\Pi(f : \mathbf{2}^{\mathbb{N}} \to \mathbb{N}).\ \Sigma(n : \mathbb{N}).\ \Pi(\alpha, \beta : \mathbf{2}^{\mathbb{N}}).\ \alpha =_n \beta \to f(\alpha) = f(\beta).$$

Proof sketch. Use the lemma by taking

$$A(n) \ :\equiv \ \Pi(\alpha, \beta : \mathbf{2}^{\mathbb{N}}).\ \alpha =_n \beta \to f(\alpha) = f(\beta).$$

Corollary. The topological topos also validates the Curry–Howard interpretation of the uniform-continuity principle for functions $\mathbf{2}^{\mathbb{N}} \to \mathbb{N}$.

C-spaces and continuous maps

Def. A C-topology on a set X is a collection P of probes $\mathbf{2}^{\mathbb{N}} \to X$ subject to the following probe axioms:

1. All constant maps are in P.
2. If $t\colon \mathbf{2}^{\mathbb{N}} \to \mathbf{2}^{\mathbb{N}}$ is uniformly continuous and $p \in P$, then $p \circ t \in P$. (Presheaf condition)
3. For any two maps $p_0, p_1 \in P$, the unique map $p\colon \mathbf{2}^{\mathbb{N}} \to X$ defined by $p(i * \alpha) = p_i(\alpha)$ is in P. (Sheaf condition)

A C-space is a set X equipped with C-topology.

A function $f\colon X \to Y$ of C-spaces is continuous if $f \circ p \in P_Y$ whenever $p \in P_X$. (Naturality condition)

Examples of C-spaces

All continuous maps from $\mathbf{2}^{\mathbb{N}}$ (with the usual topology) to any topological space X form a C-topology on X:

- Any constant map $\mathbf{2}^{\mathbb{N}} \to X$ is continuous.
- The composite $\mathbf{2}^{\mathbb{N}} \xrightarrow{t} \mathbf{2}^{\mathbb{N}} \xrightarrow{p} X$ of two continuous maps is continuous.
- The sheaf condition is satisfied because $\mathbf{2}^{\mathbb{N}}$ is compact Hausdorff.

Any continuous map of topological spaces is continuous w.r.t. the above C-topology, as composition preserves continuity.

Discrete C-spaces

Def. A map $p\colon \mathbf{2}^{\mathbb{N}} \to X$ into a set X is called locally constant iff $\exists(n : \mathbb{N}).\ \forall(\alpha, \beta : \mathbf{2}^{\mathbb{N}}).\ \alpha =_n \beta \Rightarrow f(\alpha) = f(\beta)$.

Lemma
The locally constant maps $\mathbf{2}^{\mathbb{N}} \to X$ form a C-topology which has the smallest amount of probes on X.

Def. A C-space X is discrete if all functions $X \to Y$ into any C-space Y are continuous.

Lemma
A C-space is discrete iff its probes are precisely the locally constant functions.

Def. We thus refer to the collection of all locally constant maps $\mathbf{2}^{\mathbb{N}} \to X$ as the discrete C-topology on X.

- The discrete C-topology on $\mathbf{2}$ or $\mathbb{N}$ is the set of uniformly continuous maps.
- The discrete space $\mathbf{2}$ is the coproduct of two copies of the terminal space.
- The discrete space $\mathbb{N}$ is the natural numbers object.

Yoneda Lemma and Fan functional

The monoid C of uniformly continuous $\mathbf{2}^{\mathbb{N}} \to \mathbf{2}^{\mathbb{N}}$ is a C-topology on $\mathbf{2}^{\mathbb{N}}$.

$$(\mathbf{2}^{\mathbb{N}}, \mathrm{C}) \;=\; \text{the exponential of the two discrete } \mathrm{C}\text{-spaces}$$

The Yoneda Lemma says that a map $\mathbf{2}^{\mathbb{N}} \to X$ into a C-space X is a probe iff it is continuous in the sense of the category of C-spaces.

Lemma
The exponential $\mathbb{N}^{\mathbf{2}^{\mathbb{N}}}$ is a discrete C-space.

Theorem
There is a continuous functional $\mathrm{fan}\colon \mathbb{N}^{\mathbf{2}^{\mathbb{N}}} \to \mathbb{N}$ that calculates (minimal) moduli of uniform continuity.

Modelling uniform continuity

$\mathbb{C}$-spaces provide a model of system T and dependent types:

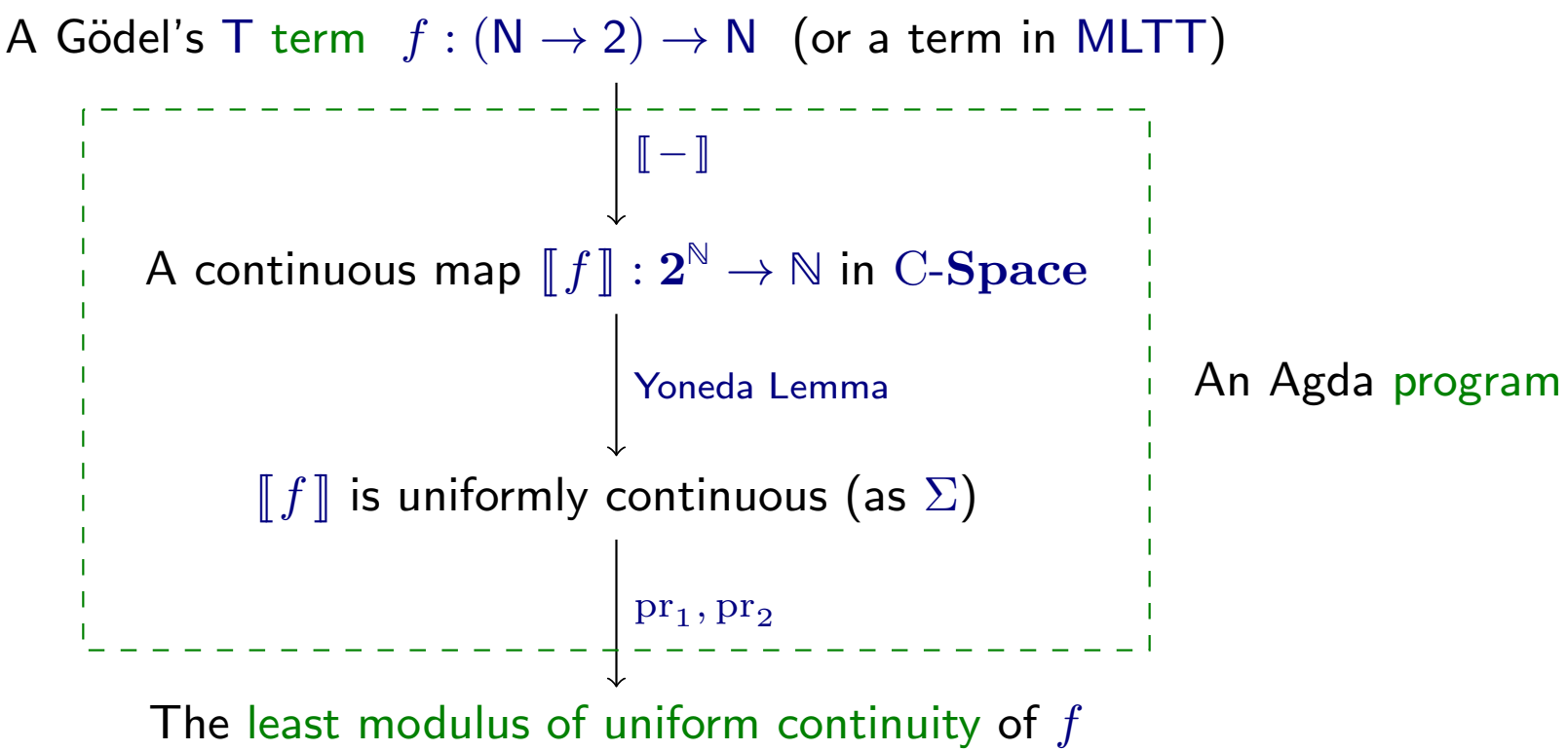
1. Cartesian closed structure — simply typed λ-calculus.
2. Locally cartesian closed structure — dependent types.
3. Natural numbers object — base type and primitive recursion principle.

Theorem
The uniform continuity axiom

$$\forall(f : \mathbf{2}^{\mathbb{N}} \to \mathbb{N}).\ \exists(n : \mathbb{N}).\ \forall(\alpha, \beta : \mathbf{2}^{\mathbb{N}}).\ \alpha =_n \beta \Rightarrow f(\alpha) = f(\beta)$$

is validated by the fan functional.

Computing moduli of uniform continuity



Summary

- The continuity principle for functions $\mathbb{N}^{\mathbb{N}} \to \mathbb{N}$ is not provable in HA^{ω}, but is validated in some models.
- The Curry-Howard interpretation of continuity is always false.
- Correct type-theoretic formulation via propositional truncation.
- For uniform continuity of $\mathbf{2}^{\mathbb{N}} \to \mathbb{N}$, the two formulations are equivalent.
- A constructive model of type theory validating uniform continuity.